

RESEARCH ARTICLE

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Stock Portfolio Management Based on AI Technology

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ABSTRACT

Forecasting stock performance is crucial for formulating a profitable trading approach aimed at achieving significant gains. In addition, prediction results serve as essential prerequisites for creating and optimizing active investment portfolios. However, predicting stock movements presents a formidable challenge due to the presence of various factors that contribute to uncertainty and instability. This paper introduces the use of a long- and short-term memory network to forecast stock movements by analyzing past data as a component to be used in portfolio optimization. To establish an effective investment portfolio, a hybrid portfolio optimization proposal is made to enhance portfolio performance while considering the diversification of assets through categories. The sensitivity of the proposed technique to the parameters is explored to understand the advantages and limitations of the different choices.

1 | Introduction

Portfolio is a set of financial assets consisting of various securities such as stocks, bonds, exchange-traded funds (ETF), real estate, cryptocurrency, alternative assets, and derivative products, held by a particular person or group (Akkaya 2021). Therefore, a portfolio can be understood as a compilation of investment assets or stock baskets, and portfolio management involves the process of making investment decisions based on customized tactical strategies to maximize returns for different investment time frames, through a cohesive investment strategy, timeline, and risk tolerance (Ta et al. 2020).

Two widely used strategies for overseeing investment portfolios are conventional and quantitative methodologies, both of which have seen the introduction of numerous techniques over the course of several decades. These strategies exhibit certain shared traits, including the examination of a restricted range of impactful variables that influence the values of the stocks, the

analysis of previous data to assess these variables, the establishment of criteria for selecting stocks, and the ongoing evaluation of performance as time progresses (Yun et al. 2020). However, traditional portfolio management is heavily based on in-depth analysis, consideration of regime changes, key characteristics, and qualitative factors. In contrast, quantitative portfolio management emphasizes exploring a wider range of investment options, maintaining discipline, verifying strategies, managing risks, and seeking lower fees. Mathematical and statistical methods constitute a dominant part among the methods (Varga-Haszonits et al. 2016). Since mean variance analysis, the first modern portfolio management method that uses a mathematical and statistical approach, has been introduced, a variety of variants of the methods have been studied (Kalayci et al. 2019). Lately, machine learning (ML)-driven techniques have been increasingly regarded as substitutes and additions to traditional statistical methods. This shift is driven by the exceptional accuracy achieved through rapid advances in both ML and computational hardware.

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Haugen and Baker (1991) concisely proclaimed that *matching the market is an inefficient investment strategy*. These authors presented one of the first empirical studies of the minimum-variance portfolio. Their contention is that even under the assumption of rational investor behavior and optimization of risk–return equilibrium in an *informationally efficient* capital market, theory indicates that cap-weighted portfolios can still be deemed inefficient investments.

Although some studies provide a framework for portfolio management based on prediction, these studies do not apply recent developments in prediction models, such as novel deep learning (DL) techniques and long- and short-term memory (LSTM) (Yun et al. 2020). Without a lack of generalization capabilities, in this work, portfolio components will be selected among stocks due to the relatively high frequency of price information available through the market, which makes it easy to optimize the involved decision-making process. Therefore, this work aims to introduce such a framework in an inductive way, based on the constructive methodology adopted.

The justification for optimization needs is evidenced because there are three types of investors in financial markets: foreign (institutional), domestic, institutional, and individual investors. Although several studies on trading performance have concluded that foreign and institutional investors perform better than individual investors (Agarwal et al. 2009; Bae et al. 2008). Cheong et al. (2017) argue that the stronger performance of foreign and institutional investors may be attributable to their access to optimal portfolios. Therefore, the application of a similar approach can significantly benefit individual investors, avoiding the use of volume levels as a significant indicator in stock markets. Whereas stock selection strategies strongly affect portfolio performance, investors aim to maximize their expected return while maintaining the risk bounded. In addition, in recent times, the strategy can also include stress preferences that regulate type of investment, such as environmental, social, and governance (ESG) principles.

For the evaluation of the methods, the indices of the stock market are commonly used. Since the indices are synthetic, they can hardly represent the actual buy/sell costs of a market. In this work, the sliding window configuration is adopted to provide a more effective way to evaluate procedures.

The subsequent sections of the paper are organized in the following manner. Section 2 presents a brief review of relevant publications and justifies the novelty of the present work. Section 3 elaborates on the proposed method, including the framework that is being proposed. Section 4 explains the results obtained according to the method and the data selected in Section 3. Finally, Section 5 discusses the results, and Section 6 concludes with the global assessment, limitations, and proposals for future research.

2 | State of the Art

As claimed in Henrique et al. (2019), due to the large number of variables that could potentially affect stock values, as well as unanticipated noise, forecasting stock markets is a difficult

job. The Efficient Market Hypothesis (EMH) Fama (1970) and the Random Walk theory Malkiel (2003) state that market returns cannot be precisely predicted. This belief is based on the presumption that investors are rational and that everyone has access to all data from the public market. The EMH has been frequently challenged, and different works by behavioral economists and econometricians (Hsu et al. 2016; Rachev et al. 2002) have shown that there is reason to question this hypothesis, as shown by the development of consistently profitable factors based on market anomalies (Azevedo and Hoegner 2022; Azevedo et al. 2023).

Price evolution is a confluence of buyers and sellers whose economic decisions are driven by expectations. These expectations are not only rational; they are also built on personal beliefs, often faced with social influence. Since the 1990s, empirical data published by behavioral finance specialists have shown that investor psychology is what drives the stock market (Daniel et al. 1998).

The capital asset pricing model (CAPM) has been subject to extensive scrutiny since its inception in the 1960s (Neslihanoglu et al. 2017). Specifically, the effectiveness of the market capitalization weighted index has been challenged, leading academics and professionals to propose various alternative investment options. Traditional models for predicting market behavior were those based on fundamental analysis (company evolution) or technical analysis (price evolution) (McMillan 2016). The ongoing introduction of new models and concepts in this domain has played a crucial role in the advancement of modern financial theories. Complex networks, now widely used as analytical tools, have evolved notably in portfolio optimization in recent times, becoming increasingly sophisticated (Guerard et al. 2015; Karathanasopoulos et al. 2017; Lin and Taamouti 2024).

In the context of optimal portfolio construction, research in the timber and forest sector demonstrates that incorporating ESG scores into investment strategies can achieve risk-adjusted returns comparable to traditional investments. This supports the feasibility of socially responsible investments in maintaining financial performance while promoting sustainability (Arreola Hernandez and Al Janabi 2020; Löff et al. 2023). Furthermore, studies in Latin America confirm that portfolios with high-ESG stocks outperform those with low-ESG or nonreporting companies, showcasing the benefits of ESG criteria in investment decision-making (Useche et al. 2023).

Portfolio management encompasses the tasks of choosing stocks to incorporate into a portfolio, determining the allocation of capital to each stock, and identifying the moments that require portfolio rebalancing. Throughout this process, it is crucial to take into account the investor's risk tolerance, as certain investors are more open to assuming higher levels of risk in hopes of achieving potentially larger returns. In this work, the authors implicitly propose adopting an active management strategy, which allows us to make decisions both on stocks and on the time to rebalance the portfolio.

In recent years, several groundbreaking contributions have emerged in the field of stock portfolio optimization,

incorporating advanced ML techniques and innovative algorithms. These contributions have significantly improved the precision and efficacy of stock market predictions and portfolio management. Academics strive to reconcile the dual objectives of maximizing returns and minimizing variance, which initially appear contradictory, in order to optimize diversified portfolios. They develop new models and estimation methods to enhance the Markowitz model. The application of network theory to portfolio research offers a fresh approach to solving asset selection challenges. In this framework, assets are viewed as nodes, and the relationships between assets form interconnected edges (Tumminello et al. 2005).

One notable advancement is the mean Value-at-Risk (VaR) model combined with AdaBoost prediction, which has demonstrated superior performance in the optimization of the multinational stock market portfolio compared to other ML regression models (Behera et al. 2023). This model improves the predictability and reliability of financial returns, providing investors with robust tools to manage multinational portfolios.

Another significant development is the use of LSTM DL models to predict future stock prices with high precision. The method has been successfully applied to different sectors of the Indian stock market, offering convenient predictions that aid in optimizing portfolios of stocks (Sen et al. 2021). Similarly, according to Jang and Seong (2023), by integrating modern portfolio theory with deep reinforcement learning, it is possible to outperform traditional algorithms in terms of Sharpe ratio (SR), annualized return, and maximum drawdown (DD) (Dacorogna et al. 1999). This combination leverages the strengths of both theoretical and practical approaches to achieve optimal portfolio performance.

However, portfolio optimization currently lacks the flexibility to create a parametric selection of stocks from diverse categories beyond their sectors, hindering the ability to assess performance under constraints related to diversification, adherence to ESG principles, and other criteria.

Therefore, this research aims to address these limitations and incorporate advanced forecast capabilities with advocating attributes such as the ability to regularly rebalance portfolio compositions. This approach seeks not only to address performance issues but also to accommodate dynamic changes in stock categorization across different categories.

3 | Data and Methodology

In this section, the authors aim to present the value proposal for the adopted approach, which is based on a proposed conceptual framework enabling splitting the problem between the stock forecasting problem and the optimization at risk one.

Taking into account the high transaction costs in practice and the limited capital and energy of investors, the number of assets included in the portfolio n is set to 3, 6, and 9, as an example. This allows us to analyze the performance of different portfolio configurations and thus select the most suitable number of investments.

In fact, the proposed optimization algorithm is introduced as well as a dataset to provide an illustrative experiment. Then, the stock forecasting problem is introduced as an uncoupled problem but critical to enable independent optimization and rebalancing strategies.

3.1 | Framework

Portfolio construction involves two main processes: asset selection and weight allocation. Regarding asset selection, Markowitz's theory states that including negatively correlated assets enhances risk diversification (Lee et al. 2016). To assign quantities to selected assets, strategies based on the minimum-variance and mean-variance models are referential approaches (Clarke et al. 2006; Goktas and Duran 2019). They are based on the compensation strategy that provides a degree of stability to the portfolio over time. Almost all approaches to optimizing a portfolio of stocks deal with these two coupled problems.

Instead, the main contribution of this research is not to select individual assets but to expand the concept of targeted asset to a category, depending on their sector or other aspects such as ESG performance. Then, the idea is to consider the variance to decide about weights, but in combination with the expected performance of these stocks, with some estimation of the computational uncertainty of the prediction, while a minimum number of categories is required to build the portfolio. This aspect is very relevant since the selection not only looks for diversification like the variance principle but also looks at considering the expected performance, which is more proactive (see Figures 1 and 2).

The proposed approach has the additional effect of enabling an uncoupled configuration, since the prediction of future evolution of stocks can be carried out at any time and with different models and techniques.

The optimization of portfolios built over families of stocks can be independently assessed, and when necessary, a rebalancing proposal can be the result of an optimization process that also considers transaction costs (see Figure 2).

3.2 | Optimization Algorithm

The optimization problem involves different subproblems. The first is related to the selection of stocks from different categories to set up the portfolio mix. The second is the rebalancing approach at discrete times according to the investment strategy selected by the owner of the portfolio. Finally, the last one, which can be done regularly and uncoupled from the two others, is the regular forecast of the different stocks potentially targeted by the owner of the portfolio since they are part of the different categories under consideration. Let us formulate these problems.

The owner of the portfolio has selected the interesting groups of stocks according to his priorities, and each group is hereinafter named a category that depends on the owner's criterion. In the following, it is assumed that there are K categories. That is, let S be the set of assets, where $\{S = \cup_{k=1}^K C_k\}$. Obviously, different

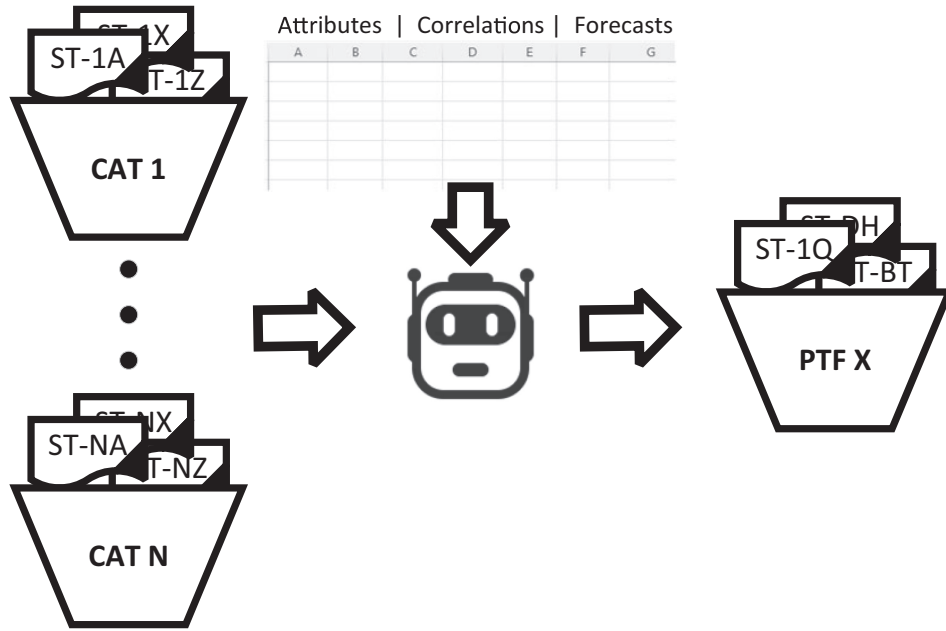


FIGURE 1 | Selection of portfolio assets considering categories, variance and expected performance.

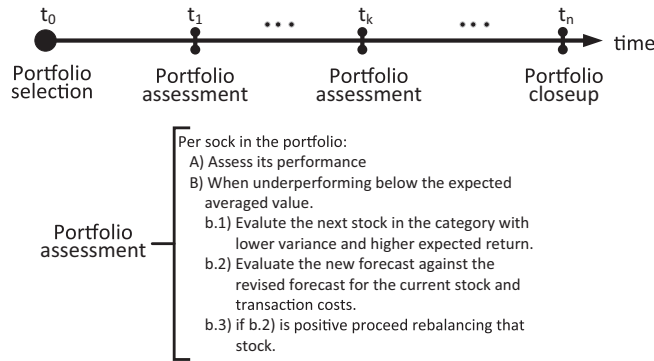


FIGURE 2 | Discrete rebalancing strategy.

assets belong to different categories C_k . Formally, let us introduce the matrix $\underline{B}$ to describe the relationship between assets and categories, such as $B_{ij} = \begin{cases} 1 & \text{if asset } i \text{ belongs to } C_j \\ 0 & \text{otherwise} \end{cases}$.

The selection of assets is based on minimum-variance and maximizing the expected return in the defined horizon of the portfolio. Finally, it takes the sum of the assets' weights to be one as a constraint and aims to find the optimal weight for the defined objectives.

$$\begin{aligned} & \bar{\omega} \min \left(\bar{\omega}^t \sum_{cov} \bar{\omega} - \lambda \cdot \sum_{i=1}^{card(S)} \omega_i r(i, t_n) \right) \\ \text{s. t. } & \begin{cases} \bar{\omega}^t \cdot \bar{1} = 1 \\ r(i, t_n) = 100 \cdot \ln \left(\frac{P_i(t_n)}{P_i(t_0)} \right) \quad \forall i \in 1, \dots, card(S) \\ \left| \underline{B\bar{\omega}} \right|_0 \geq N \end{cases} \end{aligned} \quad (1)$$

where Σ_{cov} represents the covariance matrix of the returns on assets when assets belong to different categories and one

otherwise. $card(S)$ is the cardinality of the portfolio (number of different assets); and $P_i(t)$ represents the closing price of the asset i at time t . T represents the horizon for the planned investment portfolio, and N is the minimum acceptable diversification in categories of the portfolio measured by the norm L_0 . Finally, $\lambda > 0$ is a parameter that allows one to accommodate the relevance of variance and the range of benefits.

If we consider the rebalancing subproblem, the formulation at time t_s is presented in Equation (2).

$$\begin{aligned} & \bar{\omega} \min \left(\bar{\omega}^t \sum_{cov} \bar{\omega} - \lambda \cdot \sum_{i=1}^{card(S)} \left(\omega_i^j r(i, t_n) - TC_i^j(t_s) \right) \right) \\ \text{s. t. } & \begin{cases} \bar{\omega}^t \cdot \bar{1} = 1 \\ \left| \underline{B\bar{\omega}} \right|_0 \geq N \end{cases} \end{aligned} \quad (2)$$

where ω_j refers to the weight of the asset i being replaced in the portfolio by the asset j at time t_s , where this asset still belongs to the same category k as i . The $\bar{\omega}$ represents the whole set of portfolio weights where the asset i was replaced by the asset j , both belonging to the same category, and $TC_i^j(t_s)$ represents the transition costs for the replacement of the asset j by the asset i at time t_s . ϵ represents the minimum threshold to avoid taking actions without enough motivation.

Finally, the previous algorithms are supported by the need for an independent and robust estimate of the closing prices of assets at the selected time horizons $t_s \in t_0, \dots, t_n$. To facilitate this approach, this research was intended to use cutting-edge technologies from the ML area. As discussed in the state of the art, the most promising tools are related to the field of recurrent neural networks (RNNs) by using LSTM as well as to the Transformers, where multiheading attention mechanisms can help to estimate trends based on the observation of historical

evolution observed in the past. In this case, the problem can be operated at the stock level, after observing that a single general model performed worse than individual models per asset (Candemir and Karahan 2024).

Therefore, additional effort has been made to model and understand the behaviors of these models to improve the estimation of $r(t_s)$. In 2017, a seminal paper Vaswani et al. (2017) introduced the Attention mechanism in LSTM, applied to the natural language processing (NLP) domain. On the other hand, one of the first studies to suggest the use of an LSTM + attention mechanism for multivariate prediction of time series is (Shih et al. 2019). Li et al. (2018) used a multi-input attention-LSTM to separate useful information from negatively associated elements and eliminate their noise. Qiu et al. (2020) used an attention mechanism to de-noise historical stock data. Similarly, Li et al. (2022) combined a transformer encoder and an attention mechanism together with social media *tweets* to predict stock movements. Social networks are used only to select the most relevant stocks for the study.

This experiment uses the neural network Keras front end run on Google's TensorFlow library using the Python language to perform network implementation. To evaluate the predictive power of sentiment, we consider a persistence model as a benchmark and compare its results with LSTM, stacked-LSTM, bidirectional-LSTM, CNN-LSTM, Attention-LSTM, and multivariate LSTM. We consider the predictive power through different forecasting horizons. ML algorithms with memory features extend the forecasting capacity in a daily-weekly range according to Shah et al. (2019) works. We have experimented with different numbers of neurons as part of our hyperparameter tuning process.

The aim was to find the optimal number of neurons that would result in a prediction model that is neither under- nor overfitting the data. Bhandari et al. (2022) found that 100–50–20 neurons may be the most appropriate configuration for a three-layer model, which is consistent with our work. According to Zhang et al. (2018), the number of hidden layers that minimize model error is 3. The different models use a sliding window of length 15, 25, 50, and 75 days to forecast 2, 8, 16, 30, 60, and 90 days ahead. Mean-squared error (MSE) of the models of the proposed models is calculated in the test set. All simulations have been repeated 10 times, gauging the random setup of weights and the random seed. The learning ability of the LSTM network is determined by the amount of neurons it contains. The batch size indicates how many input samples your LSTM should examine before changing the weights. Our LSTM univariate model involves 25, 50, and 75 hidden neurons in the hidden layer with a batch size of 500. Training the LSTM network involves selecting the training parameters: different window sizes: 15, 25, and 50 days. The activation function is ReLu, and the optimizer selected is ADAM, while the MSE was selected as a loss function. To produce a significant number of tests with such a diverse set of configurations, a tool has been developed and made available on GitHub <https://github.com/jordieres/Finance-AI>.

Unlike RNN, LSTM can preserve memory and state related to past activation rather than completely replacing it. They can retain characteristics for a long time thanks to the memory effect,

which also makes it possible for back-propagation to occur across a number of constrained nonlinearities, reducing the risk that the gradient will vanish and allowing the RNN to learn the long-range dependencies across time steps. The rate of adjustments determines the gradient value.

Its structure is well-known:

- The input gate controls how much of the new cell state should be retained.
- The forget gate controls how much of the current memory should be forgotten.
- The output gate controls how much of the cell state should be revealed to the network's upper levels.

In addition, to contribute to the analysis of the DL capabilities, another technique has been implemented. It is Transformers, which are a type of neural network architecture introduced in 2017 by Vaswani et al. (2017). They revolutionized the field of NLP by providing a novel approach to sequence-to-sequence tasks. Unlike traditional RNNs, transformers rely solely on self-attention mechanisms to process input sequences in parallel, making them faster and more efficient. In connection with some other works (Tang and Matteson 2021; Vaswani et al. 2017; Zeng et al. 2023), we wanted to evaluate their potential contribution to forecast stocks. The Transformer architecture uses multi-head attention, which allows the model to jointly attend to information from different representation subspaces at different positions. This enables the model to capture complex contextual relationships between different parts of the input sequence. The other two subproblems are handled by algorithms for conventional constrained minimization strategies.

The innovative idea here was to use a mechanism depending on the size of the window used to recall relevant memory length to estimate trends, enabling a variable normalization, which makes the process rather independent from monotonic tendencies and still keeps the meaning on derivatives as relevant for estimating trends. In fact, several authors have pointed out the relevance of human opinion for stock values (Swathi et al. 2022; Valle-Cruz et al. 2022; Yang et al. 2022); therefore, data from social networks related to companies were included, as well as other potentially relevant variables, to allow for multidimensional time series forecast.

3.3 | Data Description

The analysis is carried out for the top 15 NASDAQ-listed stocks presented in Table 1, by market capitalization as of December 2020, which we monitored during the period January 1, 2015 and May 30, 2024 to predict the daily open price of the stocks for a forecast horizon that includes a range of days.

NASDAQ index was selected as it is a liquid market listed under strong supervision that prevents market manipulation and has sizable components mainly from the technology industry that concentrate most of the *twitter messages* related to stocks (Mao et al. 2012), which is the basis for the hypothesis to be tested. The 15 companies listed in the top 15 on the NASDAQ account

TABLE 1 | Top 15 NASDAQ index components by market capitalization.

Ticker	Company	Index (%)
AAPL	Apple Inc.	11.38
MSFT	Microsoft Corp.	10.55
AMZN	Amazon.com Inc.	7.43
GOOG	Alphabet Inc. Class C	4.12
FB	Facebook Inc. Class A	3.96
GOOGL	Alphabet Inc. Class A	3.80
TSLA	Tesla	3.73
NVDA	Nvidia	3.63
PYPL	PayPal	2.25
ADBE	Adobe	2.16
CMCSA	Comcast	1.91
CSCO	Cisco Systems Inc.	1.72
NFLX	Netflix	1.70
PEP	PepsiCo Inc.	1.53
INTC	Intel Corp.	1.47

for approximately 61% of the total index market capitalization. In the S&P500 Index, the top 15 listed companies represent approximately 33% of the index. As the *tweets* and news are specific to the company, the higher the market share of the top 15 companies, the better the forecasting ability of the sentiment for the overall behavior of the index.

The data used in this research were historical daily open prices of shares, technical and sentiment indicators as provided by the *Bloomberg* professional platform, which accounts for professional sentiment. With the daily open price of the stock, we adjusted to account for any trend effect.

In addition to the open price and volume of the stock, we took into account the sentiment on Twitter and the news for each of the top 15 NASDAQ-listed companies.

Feature engineering or feature extraction refers to the process of transforming unstructured raw data into a vector space. This crucial phase requires expertise in the subject matter. The more problems feature extraction solves, the fewer challenges the ML system faces (Long et al. 2019). This implies that the use of ML becomes simpler and the better we are able to organize our input data from the large amount of information available because the data features used to train the ML models have a huge influence on the performance they can achieve. Sometimes, including irrelevant features in the input data can decrease the accuracy of many models and increase running time (Hira and Gillies 2015).

Then, preprocessing of sentiment information was carried out by adding four new columns to the dataset: positive and negative

ratio for Twitter/X and news sentiment analysis. For example, the positive tweet ratio was calculated as the count of positive publications divided by the total number of publications.

The second part is the calculation of the volatility. In this case, the method used to calculate the volatility of the stock has been the standard deviation of the daily percentage change in the price of the stock. Although volatility is not a raw data column, it is one of the most important metrics in market data. Volatility is used in a wide range of formulas, systems, and applications in the field of financial engineering and quantitative finance (Bhowmik and Wang 2020; Chowdhury et al. 2022). Following volatility, the next step is to calculate the trend. For this project, the price trend is computed, as well as the volume trend. The reason for this decision is because volume is a metric commonly used by traders to try to predict the market (Bordino et al. 2012). The next steps are the drop of the nans values, because the percentage of nans compared to the total data is not relevant, so it can be drop it.

- NASDAQ stocks:

For these stocks, we retrieve daily open, high, low, and close prices, volume, and Twitter and news sentiment plus RSI_{14d} technical indicator (Dash and Dash 2016). We used open prices because most of the returns are realized overnight with an open gap the next day. We calculate a trend component for price (Equation 3) and volume (Equation 4):

$$P_x^{Trend} = 2 * \frac{(P_x^{Last} - P_x^{Open})}{(P_x^{Open} + P_x^{Last})} \quad (3)$$

$$P_x^{VolTrend} = P_x^{Trend} * Volume \quad (4)$$

- Investor sentiment for those stocks are assessed through Twitter and news sentiment provided by *Bloomberg* professional platform:

Twitter sentiment. Since its launch in March 2006, Twitter has grown to become one of the most widely used social networks in the world, with 229 million daily users in 2022 (www.oberlo.com/blog/twitter-statistics). *Bloomberg* began to monitor and categorize *tweets* beginning in 2015 through its highly regarded *Bloomberg Labs* group that consistently ranks top in the annual NLP competition *SemEval*. The most mentioned companies on Twitter are mainly those that comprise the Facebook, Apple, Amazon, Netflix, and Google (FAANG).

Souza et al. (2015) defined equation (Equation 5). He calculated the relative sentiment of a company by counting how many positive ($G(t)$) and negative ($B(t)$) messages were received that day t about that company:

$$S_R(t) = \frac{G(t) - B(t)}{G(t) + B(t)} \in [-1, 1] \quad (5)$$

$S_R(t_0) = +1$, represents a day t_0 with the highest positive sentiment for the company considered. In contrast, $S_R(t_0) = -1$ indicates the highest negative sentiment, while neutrality is achieved when $S_R(t_0) = 0$.

News sentiment. As stock market participants change the way news is ingested, the tracking of event-driven news with machine reading of news is on the rise. Machines can read the news and act more quickly than a person. A 600-word piece would normally take 3 min to read because the average adult can only read 200 words at a time. These sentiment scores on a news item or *tweet* as important for a specific stock vary from -1 for negative things to 1 , with 0 being neutral. When used in trading, sentiment can serve as a directional indication to determine whether stocks in your portfolio should be held long or short. A common behavioral presumption is that if a company receives good news, its share price will increase, and vice versa. However, positive news rapidly increases stock returns, while bad news causes a long delay in response (Heston and Sinha 2016).

Regarding the Portfolio construction, different configurations have been considered, including discrete periods of 15, 30, 60, and 90 days, in coherence with the stock prediction. In addition, configurations were set for three different categories of stocks, although all of them are related to technology. The first category deals with companies producing goods and services for people's everyday working (AAPL, ADBE, MSFT, GOOG, and GOOGL); the second category focuses on services for people and entertainment (AMZN, FB, PEP, CMCSA, NFLX, and PYPL), and finally, the third one looks at hardware-centered companies (CSCO, INTC, NVDA, and TSLA). That is,

$$\underline{B} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \quad (6)$$

Such a structure in categories is just to build an example, and it does make any limitation to the proposal made in the paper. Starting portfolio life on Nov. 1st the investment with a horizon of 90 days. The design was forced to cover at least two categories and three for some experiments. The weight factor between covariance and the expected return per unit of investment has been explored in a logarithmic shape.

4 | Results

Since the Portfolio Optimization Problem deals with several subproblems (see Section 3), and most of them rely on the stock forecast, the presentation of results will begin with this one.

4.1 | Stock Prediction Subproblem

As already presented in the aforementioned Section 3 both univariate and multivariate models have been explored. This thread starts with the data windowing process, creating the new feature matrix with the window size extracted from the configuration file. Then, the data are adapted to the ahead value for the different periods' prediction. And finally, the normalization process starts; in this case, the normalization method applied is the min-max normalization, both for univariate and multivariate cases.

After preprocessing, learning of the different models was undertaken using the cross-validation strategy. The adopted error metric was the MSE. For the transformer model, the architecture included an embedding layer, as well as encoder layers with a multihead attention layer and a positional feed-forward network. Additionally, a dropout was applied after the multihead attention and feed-forward network to prevent overfitting. Dropout regularizes the model by randomly turning off neurons during training, improving its generalization ability.

Different sets of models have been built, as presented in Figure 3, where different configurations as parameters were tested to gain knowledge of the strengths and limitations of those techniques in the different assets.

Where different architectures were explored, including different memory window lengths (2, 5, 10, and 15 days) to analyze the influence of history on future evolution of the asset. In the case of LSTM, several configurations, such as pure LSTM, bidirectional-LSTM, stacked-LSTM, and attention-LSTM, were explored. For the transformer case, different models have been built, including a different number of transforming layers and number of attention heads. In fact, both one-dimensional transformers and multidimensional transformers have been modeled, where different input variables have been considered in addition to the daily closing price (P_x^{Last}), such as P_x^{Trend} , RSI_{14d} , $P_x^{VolTrend}$.

Although the purpose of this paper is not to show which configuration is the most convenient and also because behavior significantly depends on the stock itself, some rules have been derived. The transformer technology with a high number of heads has good performance for long prediction periods (see Figures 3 and 4).

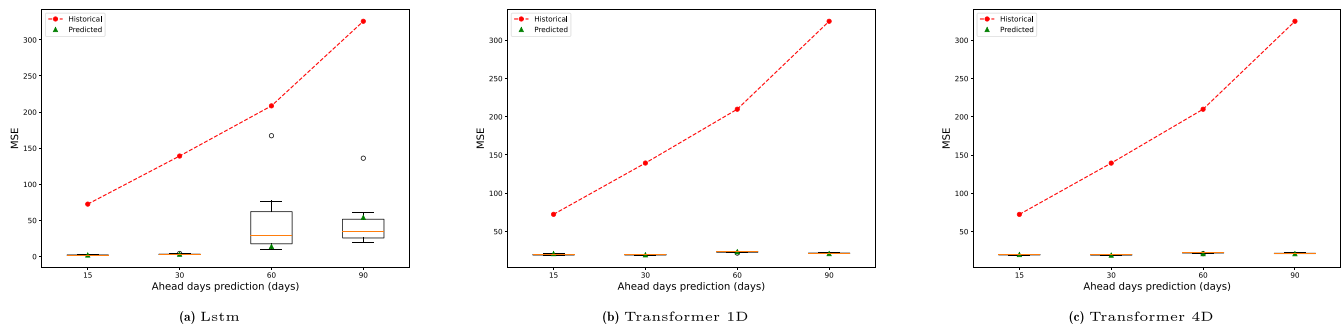


FIGURE 3 | Summary of the performance of different models with 10-fold cross-validation for GOOGLE Class C assets.



FIGURE 4 | Prediction with different time windows for the asset value of Apple stock by using a multihead attention model with 16 heads and eight layers.

4.2 | Portfolio Performance Evaluation

SR and DD have been selected for portfolio performance evaluation metrics, although several other metrics could also be adopted, such as turnover rate and break-even transaction cost.

SR measures the excess return per unit of risk. The larger the SR, the higher the risk premium per unit of risk. The SR is expressed as follows:

$$SR = \frac{E(r_p) - r_f}{\sigma_p} \quad (7)$$

where $E(r_p)$ and σ_p are the average out-of-sample return and standard deviation of the portfolio, and r_f is the risk-free rate.

DD reflects how much a portfolio's value falls from its historical peak. The DD is expressed as follows:

$$DD_i(t_n) = \sup_{s \in [0, t_n]} (\bar{P}_i(s) - \bar{P}_i(t_n)) \quad (8)$$

where $\bar{P}_i(t) = \sum_{j \in \text{Card}(S)} (P_j(t) \cdot \omega_j)$ is the value of the portfolio i on day t , with $P_j(t)$ is the predicted price of the stock j at time t .

4.3 | Portfolio Evaluation

The portfolio optimization process is performed by solving Equation (3), while rebalancing decisions are evaluated using Equation (4). Although various strategies, such as Genetic

Algorithms, can be employed for optimization, we opted for a numerical method in this small-scale experiment for simplicity. This method minimizes an objective function in a multidimensional space through a direct search, which is based on function comparison and is particularly useful for nonlinear optimization problems where derivatives are not readily available. However, the Nelder–Mead technique is a heuristic search method that can converge to nonstationary points in problems that can be solved by alternative methods (Gao and Han 2012). Larger problems may require a tuning of the technique in use, depending on circumstances. However, grouping stocks into categories significantly reduces computational effort while increasing the conceptual meaning of investment decisions to be made.

The experiments carried out are summarized in Figure 5, where each simulation corresponds to a different configuration. In Figure 5a the bar graphs are related to the left scale, as well as the predicted performance. Meanwhile, the variance and λ curves are related to the right units, represented on a natural logarithmic scale. In Figure 5b the performance portfolio evaluation metrics have been presented, operating against the right scale of the graph, while the bar graphs reflect the structure of the portfolio in terms of the categories required and finally used and are related to the left scale of the graph. In this way, the interpretation becomes clearer for the decision-making process.

5 | Discussion

The approach adopted to establish the configuration of the portfolio facilitates the consideration of different AI-based algorithms. In this work, we have implemented several RNN-based

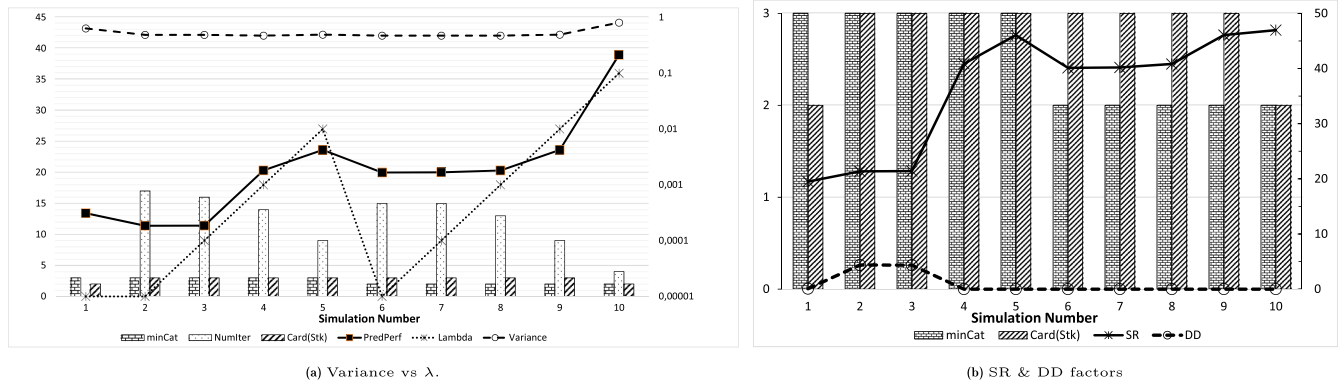


FIGURE 5 | Alternative configurations for a portfolio. Each simulation corresponds to a different configuration of a portfolio. The simulation numbers correspond to the same experiment on both Figure 5a,b.

algorithms, such as LSTM, stacked LSTM, CNN with LSTM, and attention LSTM. In addition, we have implemented different technologies, such as transformers with both individual and multidimensional dimensions and multiple attention heads. The multidimensionality involved variables such as volatility, volume of operations, technical indicator RSI, and sentiment for the asset from the general public and from specific operators, in addition to the future closing price. Such diversification in forecast algorithms brought additional insights, which evidenced that LSTM models have superior performance for short-term prediction (around 30 days), while transformers show greater stability in prediction. This behavior is not uniform across the assets, and every asset has its own behavior, which is an additional aspect where multiple modeling provides good flexibility. When multidimensionality is considered, the improvement was lower than originally expected. The main reason for this behavior is the limited amount of data available, since we operate on a daily basis and, although we started collecting data when sentiment was initially recorded, the available dataset includes 9 years and a half, but when operating in such multidimensional space, the effect *curse of dimensionality* takes place (Poggio et al. 2017).

The different simulations represented in Figure 5 correspond to different mixtures of stock selections, except the first, which is a portfolio selected by humans (baseline) where no optimization was performed. As a result, the optimization process did not require any iterations. Simulations 2 through 5 were conducted under the condition that the portfolio must contain at least three categories. In each successive simulation, the weight (λ) assigned to the expected performance relative to covariance was increased tenfold. It is evident that for very low values of λ , the variance between assets remains low. However, as λ increases, the correlation between assets increases over time. This increase in correlation can reduce the resilience of the portfolio, although it also enhances the expected performance $r_i(tn)$.

The analysis of the information derived from the same simulations and shown in Figure 5b requires one to know that the bar charts are related to the left scale and the curves are related to the right scale. It makes clear that the SR factor for portfolio configurations with a higher number of categories involved is less productive but more diverse, while configurations with two categories allow proposals involving three categories effectively involved, but SR is significantly high. On the other hand,

simulations two and three show that the DD factor is positive, which means that the highest effectiveness of the portfolio can be reached earlier than its expected life.

The hybrid approach, which integrates asset covariance and expected performance with artificial intelligence techniques, emerges as a valuable tool. It offers a knowledge-based framework that is well-suited for the portfolio configuration optimization technique employed in this paper.

The ability to assess the portfolio across different time periods and re-optimize it as needed creates a robust environment for evaluating and adjusting the chosen strategy based on empirical evidence. This approach also allows for the estimation of uncertainties associated with the assets, providing an additional layer of information that can be used to mediate decision-making.

6 | Conclusions

In coherence with Behera et al. (2023) this paper confirms the power of combining the mean VaR model and AI-based prediction of asset evolution for portfolio optimization. In fact, this paper shows that technology supports not just the definition of the portfolio but also rebalancing during its life, including reconsideration of relative importance between individual criteria. Uncoupling the forecast of asset evolution from the optimization itself enables us to use different algorithms independently. Having different estimations from different techniques allows one to implement advanced techniques such as kernel density estimation (KDE) for the application of kernel smoothing, i.e., a nonparametric method to estimate the probability density function of a random variable based on kernels as weights. By this way, indirect estimation for uncertainty can be derived, and it can be incorporated as a moderator in the selection of assets for the portfolio.

The sliding window revealed itself as a convenient approach that balances the historical time frame and variability, facilitating an agnostic understanding of the change over time. In this way, patterns can be normalized independently of the specific asset value, which facilitates learning the evolution of assets. Despite the flexibility, there is a limitation when an asset has strong variations never seen before and exceeding what has been seen

before. To mitigate such unexpected behavior, normalization preserves 5% of empty space, looking to tolerate these effects, and still having enough room to represent pattern variability.

Another significant contribution is that assets can be structured according to different criteria driven by customer preferences, making it possible to enable optimization by constraining the minimum number of categories to be considered in the portfolio. This feature aligns well with the beliefs and principles of customers, while performance is still taken into account, as well as value at risk.

More research is needed to accommodate sets of customer portfolios with a more stable and larger investment policies from larger operators, in particular when respecting specific customer beliefs, since additional opportunities can be better handled than just operating each portfolio independently. In addition, further research to extend quantum optimization to the hybrid approach presented here is something to be explored.

From a practical perspective, the adopted strategy to uncouple asset forecast and portfolio optimization significantly facilitates the computational effort, while portfolio optimization or portfolio rebalancing takes a short effort and time, becoming a task that can be included in routine management activities.

Author Contributions

The two authors have contributed equally to this article.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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